

Completion of Anaesthetic Assessments in Pregnant Patients with a Booking BMI ≥ 35 kg/m²

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Introduction

– In 2022, 10.9% of the National Women's Health patients had a maternal BMI ≥ 35 . The majority were of Pacific Peoples (47.5%) and Māori (22.5%) ethnicity (1).
– The combination of anatomical and physiological changes during pregnancy, along with the effects of obesity, results in an especially high-risk group in relation to anaesthetics (2).

Standard

– Waitematā's "Obesity in Pregnancy" guideline: 100% of antenatal women with a booking BMI ≥ 35 should receive an early referral for anaesthetic assessment.
– Process: Lead Maternity Carer (LMC) recommends completion of an anaesthetic self-assessment form as early as possible in pregnancy (3).
– Self-assessment form: Any pregnant patient with a booking BMI ≥ 35 requires automatic referral for an anaesthetic assessment by an anaesthetist.

Aim

– To investigate the completion rates of self-assessment anaesthetic forms and anaesthetic assessments conducted by anaesthetists in pregnant patients with a Booking BMI ≥ 35 kg/m².

Methods

– Retrospective audit of 150 randomly sampled patients with a Booking BMI ≥ 35 kg/m² who delivered at North Shore and Waitākere Hospitals between 1 May 2022 and 30 April 2023.
– Primary Outcome: Completion of self-assessment anaesthetic form and anaesthetic assessment conducted by an anaesthetist.

Results

Table 1: Patient Demographics

Hospital	n (%)	Age	n (%)	Booking BMI	n (%)	Ethnicity	n (%)
NSH	110 (73.3)	25-29	5 (3.3)	35-39	92 (61.3)	NZ European	71 (47.3)
WTK	40 (26.7)	30-34	103 (68.7)	40-44	34 (22.7)	Pacific Peoples	43 (28.7)
		35-39	33 (22)	45-49	18 (12)	Māori	34 (22.7)
		40+	9 (6)	50+	6 (4)	Asian	2 (1.3)

References

1. Annual Clinical report - nationalwomenshealth.adhb.govt.nz (2022) Annual Clinical Report. Available at: <https://nationalwomenshealth.adhb.govt.nz/assets/Womens-health/Documents/ACR/Reports/2022-Annual-Clinical-Report.pdf> (Accessed: 3 September 2023).
2. Mossie, A., Ali, S.A., and Tesema, H.G., 2022. Anaesthetic implications of morbid obesity during pregnancy: a literature based review. International Journal of Surgery Open, 40, p.100444.
3. Director of Midwifery (2021) 'Obesity in Pregnancy'. Te Whatu Ora Health New Zealand Waitematā.

Figure 1: Completion of Anaesthetic Assessments in Pregnant Patients with a Booking BMI ≥ 35 kg/m² at Waitematā

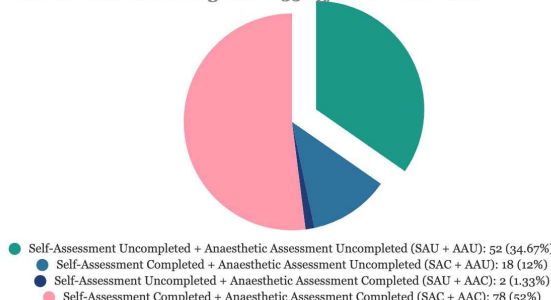
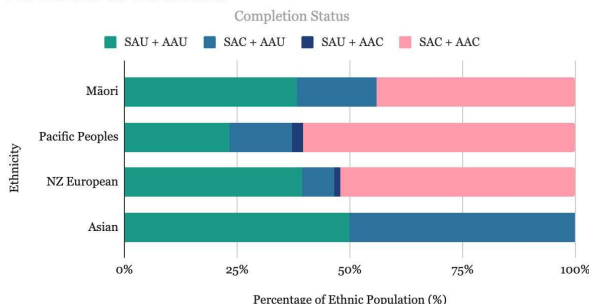


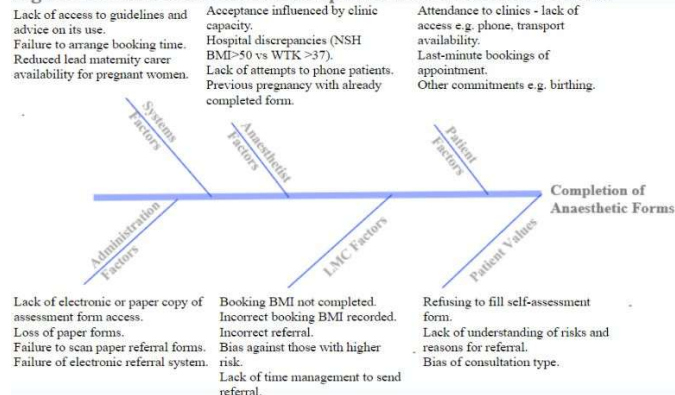
Figure 2: Completion of Anaesthetic Assessments between Ethnicities at Waitematā



Interpretation of Findings

– Approximately a half completed both anaesthetic assessments and almost a third had no form of assessment at all. Upon discussion with the Anaesthetic Department, this is a result of their clinic's reduced capacity and subsequent ability to only assess patients with the highest risks (those with higher BMIs).
– After excluding the Asian population due to their small sample size, the Māori population had the lowest rate of completed anaesthetic assessments (44%). Therefore, the Māori population is disproportionately affected by increased anaesthetic risk, causing worse health outcomes in this population.
– The following barriers can be addressed:

Figure 3: Possible Causes of Incomplete Anaesthetic Assessments



Recommendations

– According to the Waitematā's Anaesthetic Department, there may be an overstated risk associated with the impact of high Booking BMIs on anaesthetic complications. Combined with their reduced clinic capacities, it seems that an update to Waitematā's "Obesity in Pregnancy" guideline is warranted. Increasing the BMI threshold to ≥ 40 for an anaesthetic referral may not only enhance adherence rates to the guidelines but also prioritise those at the highest risk for assessment first.